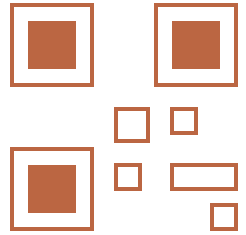
AROGYASETU · PATIENT CASE-TAKING SOFTWARE FOR INDIAN OPDS

The doctor should listen. The software should write.

From a seven-minute verbal complaint in any Indian language — to a structured, coded, shareable medical record — in a single OPD visit, driven by one QR code. Bilingual by design: Ayurveda & Allopathy in one record.



01 · THE PROBLEM, IN THE REAL WORLD

The consultation is the most valuable two minutes in Indian healthcare. Half of it is spent writing.

~2 min

Average doctor–patient consultation time in India — among the shortest of 67 countries studied

1 : 834

Government-reported doctor : population ratio — a doctor for every 834 people, ~80% of whom never reach a hospital record system

5×

The same history is retold at five desks — reception, nurse, doctor, lab, pharmacy — because nothing carries forward

Paper

Prescriptions, lab reports and discharge summaries live in plastic bags; a returning patient is, digitally, a stranger

Sources: Irving et al., *BMJ Open* 2017 (international consultation-time study, 67 countries, 2012–13 NCDRHS India data) · PIB / Ministry of Health & Family Welfare 2024 (doctor availability) · field observation at Indian OPDs.

02 · THE BARRIER NOBODY BUILDS FOR

Healthcare in India speaks a language many patients don't read.

A tribal patient in Chamarajanagar can explain their illness fluently — but cannot fill an English intake form. Studies at Indian district hospitals document patients leaving OPDs without understanding their diagnosis or medicines. Forms exclude; voice includes.

22

Scheduled languages, 100+ major mother tongues — India has no single "national" clinical language

~40%

Of Indian adults cannot read a standard prescription label comfortably

22

Languages our speech pipeline understands (Sarvam Saaras v3) — the patient speaks; the record is written for the doctor

Sources: Census of India (scheduled languages) · UNESCO Institute for Statistics, India literacy profile · Karanth & Vaid, language barriers in Karnataka district-hospital care, 2022.

03 · WHY NOW

The rails are already laid — India built the world's largest digital health network.

73.9 crore
ABHA digital health accounts created under Ayushman Bharat Digital Mission

49 crore
Health records already linked to ABHA — but most OPD software still can't feed them structured data

50 crore+
Teleconsultations on eSanjeevani — proof that Indians will use digital health at population scale

$43.4 Bn
AYUSH & herbal market size (2024) — 3,000+ government Ayush hospitals with no modern case-taking software

The infrastructure (identity, consent, records) exists. What's missing is the **point-of-care layer**: software that captures the consultation itself. That is ArogyaSetu.

Sources: PIB, Ministry of Health & Family Welfare — ABDM milestones (Feb 2025); eSanjeevani 5-crore+ milestone updates (2025–26); IMARC India AYUSH & Herbal Market Report 2024.

04 · THE SOLUTION

ArogyaSetu — one QR code, one conversation, one complete medical record.

One QR, whole hospital

Booking → check-in → vitals → consult → pharmacy → billing → discharge. Every desk scans the same code; the history is never retold.

The patient talks, the app writes

Voice-first intake and ambient scribe convert speech in 22 languages into SOCRATES-structured notes and SOAP drafts the doctor approves.

Ayurveda is computable

Prakriti, Ashtavidha and Dashavidha assessments, plus 3-way dual coding: NAMASTE ↔ ICD-11 TM2 ↔ ICD-11 biomedicine.

ABDM-ready by design

ABHA linking, FHIR R4 document bundles and care-context references for every encounter — exportable to the national grid.

8 role-based portals · patient pre-consult intake · encrypted medical Drive · family case files · referral dispatch · drug-interaction safety · FHIR export

05 · THE HERO FLOW

Ten steps, five desks, one code.
Nothing is re-entered, ever.

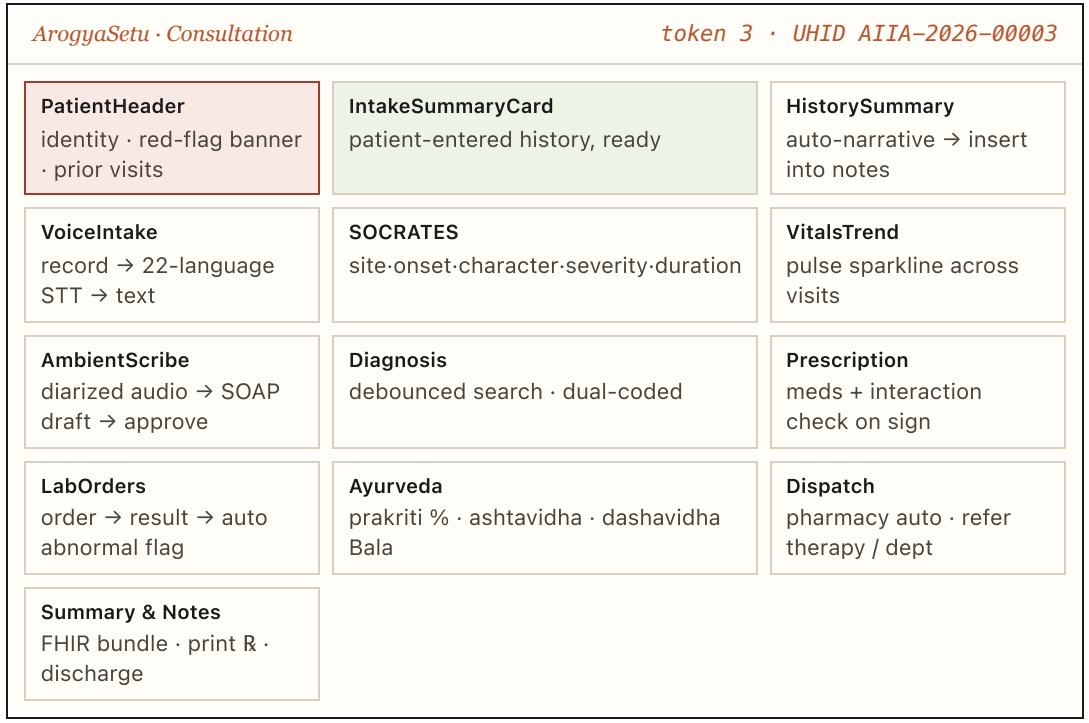
01 **PATIENT** Books appointment in 6 guided steps	02 **PATIENT** Gets QR slip `AIIA–CK–…` + queue token	03 **PATIENT** Pre-consult intake by voice, consent-gated	04 **RECEPTION** Scans QR → checked in (the only check-in path)	05 **NURSE** Scans QR → vitals recorded
06 **DOCTOR** Queue auto-promotes red-flag / emergency cases	07 **DOCTOR** Consultation screen — intake already filled	08 **PHARMACY** Scans QR → dispenses, stock + bill auto-raised	09 **BILLING** One click pay — consult + meds + labs	10 **DOCTOR** Discharge → FHIR bundle + follow-up booked

Row-locked token allocation (no double-booking) · forged or guessed codes return **404** · walk-in & emergency patients enter at the front of the queue with token E

06 · INSIDE THE CONSULTATION

Thirteen cards. Everything the doctor needs, nothing they had to type.

- **Intake arrives pre-filled** — complaints, SOCRATES history, allergies, medicines, past history, positive ROS — collected from the patient before they left home
- **Red-flag safety** — "chest pain / breathlessness / bleeding" in the complaint auto-flags the case and promotes it in the queue
- **One pick, three codes** — choosing *Amlapitta* records NAM-AYU-0001 + TM26.SA00 + K29.7 simultaneously
- **Sign with confidence** — prescriptions are interaction-checked against a severity pair-list before signing



07 · THE AI THAT DOES THE TYPING

Voice in, structure out.

Patient speaks
"Do din se seene mein jalan hai, khane ke baad badh jaati hai..."

→

Sarvam Saaras v3 STT
22 Indic languages, diarized two-speaker consultation audio

→

SOCRATES extractor
Site · Onset · Character · Radiation · Associations · Timing · Severity · Duration

→

SOAP draft
LLM drafts Subjective / Objective / Assessment / Plan

→

Doctor approves
One click — the doctor remains the signer of record

Pre-consult intake (patient, at home): consent-gated adaptive questions — the branching engine scores complaint keywords and asks only what matters (cardiac ≠ dermatology).

In the room (doctor): ambient scribe records the conversation, separates the two speakers, drafts the note; vitals trend and prior visits are one glance away.

Paper → digital: patients photograph old lab reports / prescriptions; OCR + extraction recovers values, flags abnormals against reference ranges, and the doctor confirms.

08 · THE PRINCIPLE JUDGES REMEMBER

Honest AI — the system never invents a medical fact.

1 · Real — Sarvam AI
Primary engine per capability (STT, translate, chat, TTS, OCR). Key held encrypted, admin-configurable at runtime.

→

2 · Fallback — local
openai-whisper and Ollama run on the same machine. Works offline; no cloud dependency for core capture.

→

3 · Labelled mock
Only if both fail. Response is flagged `mocked:true` with the real error attached. Never fabricates text.

- **Tested honesty:** the suite includes an explicit assertion that when no engine can transcribe, the API returns `provider:"none"` + error — not a made-up transcript
- **Admin transparency:** a live real-vs-mocked table on the admin portal shows exactly which capabilities are live
- **Mocked by design (and labelled):** ABDM sandbox bridge (OTP 123456), payment gateway
- **Human stays in charge:** AI drafts; the doctor reads, edits, and approves every note and prescription

09 · MAKING 3,000 YEARS COMPUTABLE

One diagnosis, three codes — Ayurveda and biomedicine in the same record.

Doctor picks the Ayurvedic term

Debounced search over 42 curated NAMASTE terms with synonyms (Amlapitta, Jwara, Shwasa, Kasa, Prameha, Sandhivata…)

→

3 rows written at once

NAMASTE `NAM-AYU-0001` · ICD-11 TM2 `TM26.SA00` · ICD-11 biomedicine `K29.7` (reflux oesophagitis / gastritis)

→

FHIR R4 bundle

Dual-coded Conditions + prakriti Observations in a document Bundle — legible to any ABDM-participating hospital

Prakriti & Vikriti

18-trait questionnaire scores Vata / Pitta / Kapha percentages; current symptoms identify the aggravated dosha.

Ashtavidha & Dashavidha

The 8 classical examinations (pulse, tongue, urine…) and Charaka's ten-fold assessment with a computed Bala (strength) score.

Why it matters

India has 3,000+ Ayush hospitals and 28,000+ dispensaries whose records are invisible to the biomedical system. Dual coding ends that silo.

10 · PRIVACY IS THE PRODUCT

Encrypted, consented, audited — and the audio deletes itself.

AES · Fernet

Aadhaar numbers encrypted at rest, masked XXXX–XXXX–1234 in every API response. Medical Drive blobs stored encrypted, never publicly served.

Rotating sessions

Short-lived JWT + httpOnly refresh cookie that rotates on every use; patients can see and revoke all active sessions.

Granular consent

Consultation recording requires explicit per-item consent, is revocable, and raw audio auto-purges after the retention window (default 30 days).

Audit & ownership

Every sensitive action logged; patients can read only their own records (403-tested), staff act only within their role.

Production mode refuses to boot with a development secret or missing encryption key · drug-interaction pair list guards prescriptions · access to another patient's Drive documents requires an explicit, expiring, reason-tagged grant

11 · UNDER THE HOOD

Two clean services, standard protocols, nothing exotic to deploy.

System architecture — *localhost:3000 / :8000*

Next.js 14 · React 18 · Tailwind · zustand
34 routes, 8 role portals · jsQR camera scanning · QR generation · 6 UI languages · dark mode · 401→silent refresh

FastAPI · SQLAlchemy 2 async · Pydantic v2
128 endpoints · 11 routers · SQLite (Postgres-ready) · lifespan auto-migration & seed · Swagger at /docs

Domain services
SOCRATES NLP · prakriti / dashavidha scoring · terminology 3-way mapping · adaptive question engine · OCR & lab extraction · FHIR R4 builder + validator · history summariser

AI chain
Sarvam AI (STT·translate·chat·TTS·OCR) → Whisper / Ollama local → labelled mock

Data
39 tables · encrypted blob store `media/drive/{uuid}.bin` · audit log · retention engine

- **128 API endpoints**, every one role-guarded and covered by an interactive Swagger UI — judges can poke the API themselves
- **33 automated tests** exercise the full 8-role journey, cross-desk data flow, security guards and AI honesty
- **Docker Compose** included; SQLite today, Postgres-ready ORM for scale
- **Open standards only** — FHIR R4, JWT, OAuth-style refresh, ABHA/ABDM patterns. No proprietary lock-in.

12 · BY THE NUMBERS

Built end-to-end, not a mock-up.

39 database tables	**128** API endpoints	**34** frontend routes	**33** automated tests, all passing	**8** role-based portals	**22** speech-to-text languages

42 curated terminology entries → **126** coded rows across three systems

13 interactive cards on the consultation screen · **6** UI languages

2 real AI engines + honest fallback chain · **1** QR code for the entire visit

13 · REAL-WORLD IMPACT

Where this goes next — the point of the whole exercise.

Primary Health Centres

A typical PHC doctor sees 100+ patients a day with no digital record. ArogyaSetu turns each visit into structured, coded, FHIR-exportable data — feeding ABHA, not plastic bags.

Ayushman Arogya Mandirs

India's 1.5+ lakh Ayush health & wellness centres deliver integrative care that no EHR represents today. Dual-coded records make AYUSH outcomes measurable nationally.

Continuity of care

A migrant worker treated in Bengaluru walks into a PHC in Bihar; the case file, allergies and medicines follow the ABHA account, not the paper.

Public-health signal

When every consultation is coded (ICD-11 / NAMASTE), disease surveillance moves from annual surveys to near-real-time — this is the A, of ABDM's own "Arogya Setu" spirit.

Alignment: UN SDG-3 (good health & well-being) · Ayushman Bharat Digital Mission · National AYUSH Mission · Digital India. Deployable as a single container per facility; Sarvam-first with fully offline Whisper/Ollama mode for low-connectivity districts.

14 · LIVE DEMO

Everything shown is running on this machine, right now.

- **Patient portal** — `localhost:3000/login` · patient@demo.in
- **Staff portal** — `localhost:3000/staff/login` · doctor-allo@demo.in (all 8 roles one-click)
- **API playground** — `localhost:8000/docs` · 128 endpoints, live
- **Source code** — `github.com/venkaraddi-crypto/qriuos` · 33 tests: `pytest tests/ -q`
- **Password for every demo account** — `password123`

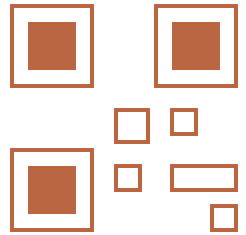
7-minute demo path

1. Patient books → QR slip (30s)
2. Voice intake, consent-gated (90s)
3. Reception scans QR (30s)
4. Nurse vitals (20s)
5. Doctor: queue → consult → voice → SOCRATES → Amlapitta → 3 codes → prescribe → FHIR (3min)
6. Pharmacy dispense → Billing pay (45s)
7. Patient sees records + case file (30s)

AROGYASETU · आरोग्यसेतु — "THE BRIDGE TO HEALTH"

The record should be the bridge, not the barrier.

One QR code connects every desk. One conversation fills the record. Two systems of medicine share one language of codes. And the doctor — finally — just listens.



Thank you · github.com/venkaraddi-crypto/qriuos · live demo available now